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1.3 Maximum Angular Displacement And Angular Velocity

The torsional motion of the driven gear is governed by

$J_o \ddot{\theta} + k_t \theta = M_t(t)$

Given Parameter :

- $J_o = 0{,}10 \ kg.m^2$
- $G = 79{,}3 \ GPa = 79{,}3 \times 10^9 \ Pa$
- $d = 0{,}050 \ m$
- $L = 1{,}00 \ m$
- $M = 1000 \ N.m$

The polar second moment of area of the solid shaft

$$J_p = \frac{\pi d^4}{32}$$

$$J_p = \frac{\pi (0{,}05)^4}{32}$$

$$J_p = 6{,}136 \times 10^{-7} \ m^4$$

The torsional Stiffness

$$k_t = \frac{G J_p}{L}$$

$$k_t = \frac{(79{,}3 \times 10^9)(6{,}136 \times 10^{-7})}{1}$$

$$k_t = 48\,657{,}87 \ N.m/rad$$

The initial steady-state angular displacement

$$\theta_0 = \frac{M}{k_t} = \frac{1000}{48657{,}87}$$

$$\theta_0 = 0{,}02055 \ rad$$

Using the broken-tooth torque disturbance in the Simulink model, the simulated response gives :

Maximum angular displacement :
$\theta_{max} \approx 0{,}0233$ rad

Maximum angular velocity :
$|\dot{\theta}|_{max} = 1{,}92$ rad/s

1.4 Dominant Torsional Vibration Frequency and Resonance

The natural angular frequency of the torsional system

$$\omega_n = \sqrt{\frac{K_t}{J_0}}$$

$$\omega_n = \sqrt{\frac{48657{,}87}{0{,}10}}$$

$\omega_n = 697{,}55$ rad/s

Convert natural frequency to Hz

$$f_n = \frac{\omega_n}{2\pi} = \frac{697{,}55}{2\pi}$$

$f_n = 111{,}02$ Hz

Rotational frequency

$$f_r = \frac{1000}{60}$$

$$f_r = 16{,}67 \text{ Hz}$$

Harmonic	Frequency (Hz)
1	16,67
2	33,33
3	50.00
4	66,67
5	83,33
6	100,00
7	116,67

~~The 7th harmonic~~

$\cancel{f_7 = 7(116{,}67)}$

$\cancel{f_7 = 116,}$

The 7th harmonic

$$f_7 = 7(16{,}67)$$

$$f_7 = 116{,}67 \text{ Hz}$$

$$\Delta f = 116{,}67 - 111{,}02$$

$$\Delta f = 5{,}67 \text{ Hz}$$